# Vinh Nguyen

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Melbourne, VIC | 045-5678-195 | Full Working Rights

## Summary

AI Engineer with an M.S. in Artificial Intelligence from UT Austin, specializing in LLM fine-tuning, RAG systems, and Deep Learning. Experienced in building end-to-end ML pipelines with MLOps tooling (DVC, MLflow, Docker), backed by a strong foundation in data analytics and operations across global organizations including adidas and New Balance, as well as start-up and family business settings.

## Education

**University of Texas at Austin** — Austin, TX
*Master of Science in Artificial Intelligence • GPA: 3.93/4.0 • ΦΚΦ Honor Society* — *Aug. 2024 – May 2026*

**Centennial College** — Toronto, ON
*Advanced Diploma in AI Software Eng Tech • GPA: 4.43/4.5 • CCSAI Scholarships* — *Aug. 2022 – Aug. 2024*

**FPT University - Aptech** — Ho Chi Minh, VN
*Diploma in Management Information Systems • GPA: 8.8/10* — *Aug. 2020 – Nov. 2021*

**Foreign Trade University** — Ho Chi Minh, VN
*Bachelor in Economics*

## Experience

**Machine Learning Student Researcher** — Jul. 2024 – Aug. 2024
*Centennial College* — *Toronto, ON*

- Led a 7-week exploratory ML research project on the newly released Open Oximeter Repository dataset; utilized SQL and data wrangling techniques to analyze data quality, validate availability with stakeholders, and assess opportunities for future $SpO_2$ estimation model development. [**Project Report**]

**Data Analyst (Intern)** — Jan. 2023 – Aug. 2024
*New Balance Canada* — *Mississauga, ON*

- Built automated reporting tools using Excel VBA, PowerQuery, and PowerPivot, converting multi-hour manual processes into sub-10-minute automated pipelines for Sales, Accounting and Planning; modified US regional dashboards in Spotfire to enable cross-border access to business reports for the Canadian teams.

**Process and Technology Analyst** — Apr. 2021 – Aug. 2022
*New Balance Vietnam* — *Ho Chi Minh, VN*

- Collaborated with global product and costing teams to analyze SOPs, optimize production workflows, and develop calculation tools, and cleared a backlog of over 300 pending issues while improving operational visibility and earning the New Balance Associate Appreciation Award.

**Strategic Project Manager** — Sep. 2019 – Sep. 2020
*HEAD Phat Thinh (Honda Exclusive Authorized Dealers - Family business)* — *Ho Chi Minh, VN*

- Led a small team to launch a centralized business management system, transitioning operations from manual Excel-based tracking to a unified platform across 6 showrooms while expanding e-commerce channels, standardizing workflows, and optimizing staffing allocation based on business needs.

**Operations Manager** — May. 2018 – Aug. 2019
*Speedia Footwear (Start-up)* — *Ho Chi Minh, VN*

- Built and managed a cross-functional sourcing and operations team spanning product development, design coordination, costing, material planning, molding, factory follow-up, and supplier communication for an early-stage footwear start-up from concept to production preparation.

**Costing Specialist** — Feb. 2012 – May. 2018
*adidas Sourcing Ltd. - Liaison Office* — *Ho Chi Minh, VN*

- Managed product costing and enterprise costing systems to support cost optimization, production milestone tracking, process improvement, and operational compliance audits while serving as a key user and trainer for costing and product lifecycle management platforms including PLM, SATRA, ShoesCost, and ProCost.

## Projects

**Myth-GraphRAG: Knowledge Graph for Literary QA** | *Python, ChromaDB, Neo4j, Cypher, OpenAI* • **Code**

- Built a GraphRAG system for The Odyssey by constructing a knowledge graph of entities and relationships, and evaluated BaseRAG (vector search), GraphRAG Global Search, and GraphRAG Local Search on 20 benchmark questions using relevance, completeness, and faithfulness metrics.
- Demonstrated that GraphRAG Local Search achieved the strongest overall performance, winning 75% of evaluations and outperforming BaseRAG (50%) and GraphRAG Global Search (25%).

**Multi-Document RAG Assistant** | *Python, LangChain, ChromaDB, HF, Streamlit* **Demo** • **Code**

- Developed a RAG system capable of ingesting PDF, DOCX, and TXT files; implemented text chunking, embedding generation, and semantic vector search using ChromaDB to enable precise, context-aware conversational QA.

**LLM Tool-Calling Weather Assistant** | *Python, Groq, Geopy, Open-Meteo, Streamlit* **Demo** • **Code**

- Engineered a deterministic tool-calling workflow utilizing Groq API to extract structured parameters from natural language prompts, seamlessly orchestrating Geopy and Open-Meteo APIs for real-time weather data retrieval.

**Grade School Math Reasoning App** | *HF Transformers, LoRA, PEFT, DVC, Gradio* **Demo** • **Code**

- Fine-tuned small LLMs on GSM8K using LoRA-based supervised fine-tuning and evaluated instructional few-shot Chain-of-Thought (CoT) prompting, Direct Answer SFT, and CoT SFT reasoning strategies.

**Restaurant Review Sentiment Classification** | *PyTorch, HF Transformers, DVC, Gradio* **Demo** • **Code**

- Fine-tuned DistilBERT for 5-class sentiment classification on the Yelp Review Full dataset, implementing context window filtering to optimize sequence length; engineered an end-to-end training pipeline with DVC.

**Quantization and LoRA Techniques** | *Python, PyTorch, LoRA, Quantization* • **Code**

- Developed from-scratch implementations of FP16/BF16 half-precision training, 4-bit quantization, and LoRA adapters for parameter-efficient LLM fine-tuning.

**Scene Text Recognition Pipeline** | *PyTorch, YOLOv11, Ultralytics, OpenCV, DVC* • **Code**

- Architected a two-stage Scene Text Recognition pipeline deploying YOLOv11 for text detection and a custom CRNN with CTC loss for sequence recognition, achieving a 98% word-level accuracy on the ICDAR 2013 dataset.

**Brain Tumor Segmentation** | *PyTorch, UNET, ResNet, Segmentation, IoU* • **Code**

- Built a UNET model with a ResNet-34 encoder trained on multi-modal MRI scans to automatically segment brain tumors, achieving an IoU score of 0.65 on the validation set.

**Self-Supervised Contrastive Learning** | *PyTorch, ResNet, Contrastive Learning* • **Code**

- Built a ResNet-18–based contrastive learning model for chest X-ray pneumonia classification, achieving 81.25% accuracy with limited labeled data.

**MLOps Pipeline System** | *Python, PyTorch, DVC, Mlflow, Poetry, Docker* • **Code**

- Built an end-to-end reproducible ML pipeline using Hydra, DVC, MLflow, Poetry, and Docker for experiment tracking, configuration management, and workflow orchestration.

**MLflow Tracking Server on AWS** | *Python, Mlflow, AWS EC2 , Amazon S3, Scikit-learn, Pandas* • **Code**

- Built an MLflow Tracking Server using AWS EC2, and S3 to enable remote experiment tracking, and artifact storage from local training workflows.

## Technical Skills

**GenAI & LLMs:** HF Transformers, Fine-tuning, PEFT (LoRA), Quantization, GRPO, RLVR, RAG, GraphRAG, Agentic AI, LangChain, MCP, Neo4j
**Machine Learning:** Python, PyTorch, Scikit-learn, YOLO, NumPy, Pandas, Matplotlib
**Data Engineering:** SQL, MongoDB, ChromaDB, PowerBI, PowerQuery, PowerPivot, Tableau, Spotfire
**MLOps & Cloud:** Docker, Git, GitHub Actions, CI/CD, MLflow, DVC, Hydra, FastAPI, AWS, Streamlit, Gradio